

HAMMARLAND MARKET, FinlandWMO: **029930**Lat: **60.301N**Lon: **19.131E**Elev: **15**StdP: **101.14**Time Zone: **2.00 (EUE)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.4	-6.8	-13.3	1.2	-8.1	-11.1	1.5	-6.3	20.1	0.5	18.1	-0.2	8.1	100	0.893

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.5	21.7	19.6	20.4	18.3	19.2	17.1	20.3	21.5	18.7	20.0	17.7	19.0	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	21.1	18.2	13.1	19.6	17.1	12.2	18.6	58.3	21.6	53.2	20.1	50.0	19.1	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.9	14.4	-10.0	22.9	4.2	1.8	-13.1	24.2	-15.5	25.2	-17.9	26.2	-21.0	27.5		
(5)			-11.2	19.9	4.1	1.8	-14.2	21.2	-16.6	22.2	-18.9	23.3	-21.9	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-0.2	-1.2	0.4	3.3	7.2	11.8	16.3	16.7	13.4	8.4	4.5	2.3
	DBStd	6.65	2.92	3.64	2.85	2.12	2.46	2.46	2.39	1.91	2.08	2.31	2.66	2.59
	HDD10.0	1699	317	312	299	202	92	11	0	0	2	60	164	240
	HDD18.3	4169	576	546	557	452	344	196	72	57	149	308	414	498
	CDD10.0	589	0	0	0	0	6	64	197	208	103	11	0	0
	CDD18.3	17	0	0	0	0	0	0	10	7	0	0	0	0
	CDH23.3	6	0	0	0	0	0	0	6	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.8	9.4	8.3	7.4	6.8	6.7	6.6	6.2	6.8	7.5	8.5	9.1	9.8
Precipitation	PrecAvg	546	44	32	24	28	37	51	49	61	57	59	57	49
	PrecMax	784	86	78	62	67	86	94	133	168	121	142	141	96
	PrecMin	417	1	3	0	4	3	13	4	3	7	0	10	5
	PrecStd	81	24	17	16	17	23	22	37	38	31	32	28	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	5.6	7.6	10.6	14.6	18.5	24.0	22.4	18.7	14.4	10.1	9.1
		MCWB	4.8	3.6	4.8	6.8	10.3	14.1	20.1	20.4	16.3	13.0	9.2	7.6
	2%	DB	5.1	4.2	5.8	8.4	12.8	17.2	22.3	21.3	17.7	13.1	9.2	7.2
		MCWB	4.1	2.7	3.5	5.7	9.6	14.1	20.1	19.2	15.8	11.8	8.4	5.7
	5%	DB	4.4	3.4	4.6	7.0	11.5	16.1	21.0	20.2	16.8	12.2	8.6	6.3
		MCWB	3.5	2.2	2.9	4.9	9.1	13.5	19.2	18.0	15.1	11.1	7.8	5.1
	10%	DB	3.6	2.6	3.6	6.0	10.4	15.0	19.7	19.2	16.0	11.6	7.8	5.5
		MCWB	2.8	1.7	2.3	4.2	8.6	13.0	17.8	17.2	14.4	10.4	7.0	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	4.1	5.2	7.4	11.7	15.9	21.8	21.7	17.4	13.4	9.6	7.9
		MCDB	5.8	5.2	7.3	9.7	14.0	17.3	22.8	22.0	18.0	13.9	9.8	9.0
	2%	WB	4.4	3.1	4.0	6.0	10.2	14.7	20.7	19.9	16.2	12.1	8.7	6.2
		MCDB	4.8	4.0	5.6	7.7	11.8	16.3	21.9	21.1	17.3	12.8	9.1	6.9
	5%	WB	3.7	2.5	3.2	5.2	9.5	14.0	19.6	18.5	15.4	11.5	8.0	5.4
		MCDB	4.2	3.3	4.3	6.6	11.0	15.4	21.1	19.7	16.5	12.2	8.6	6.1
	10%	WB	2.9	2.0	2.6	4.4	8.7	13.2	18.1	17.8	14.7	10.7	7.2	4.6
		MCDB	3.5	2.6	3.5	5.7	10.2	14.7	19.6	19.0	15.8	11.5	7.8	5.4
Mean Daily Temperature Range	5% DB	MDBR	2.6	2.6	2.6	2.9	3.2	3.1	2.5	2.5	2.4	2.2	2.4	2.6
		MCDBR	2.9	2.9	3.9	5.0	5.2	4.8	3.7	3.2	3.0	2.6	2.5	3.2
		MCWBR	2.8	2.5	2.9	3.0	3.0	2.8	2.7	2.5	2.4	2.6	2.4	3.0
	5% WB	MCDBR	2.8	2.8	3.6	4.8	4.7	4.1	3.2	2.5	2.5	2.6	2.3	3.0
		MCWBR	2.8	2.4	2.8	3.2	3.1	2.8	2.8	2.4	2.3	2.7	2.5	3.0
Clear-Sky Solar Irradiance	taub	0.274	0.313	0.332	0.368	0.367	0.358	0.385	0.375	0.345	0.331	0.297	0.382	
	taud	2.199	2.277	2.341	2.343	2.419	2.487	2.432	2.450	2.520	2.421	2.230	2.654	
	Ebn at Noon	490	679	796	829	859	874	836	816	784	659	449	345	
	Edh at Noon	40	67	86	103	103	98	101	92	73	58	39	28	
All-Sky Solar Radiation	RadAvg	0.24	0.80	2.31	3.96	5.37	6.01	5.59	4.35	2.70	1.16	0.33	0.15	
	RadStd	0.03	0.16	0.25	0.34	0.43	0.39	0.54	0.38	0.28	0.18	0.08	0.02	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page